

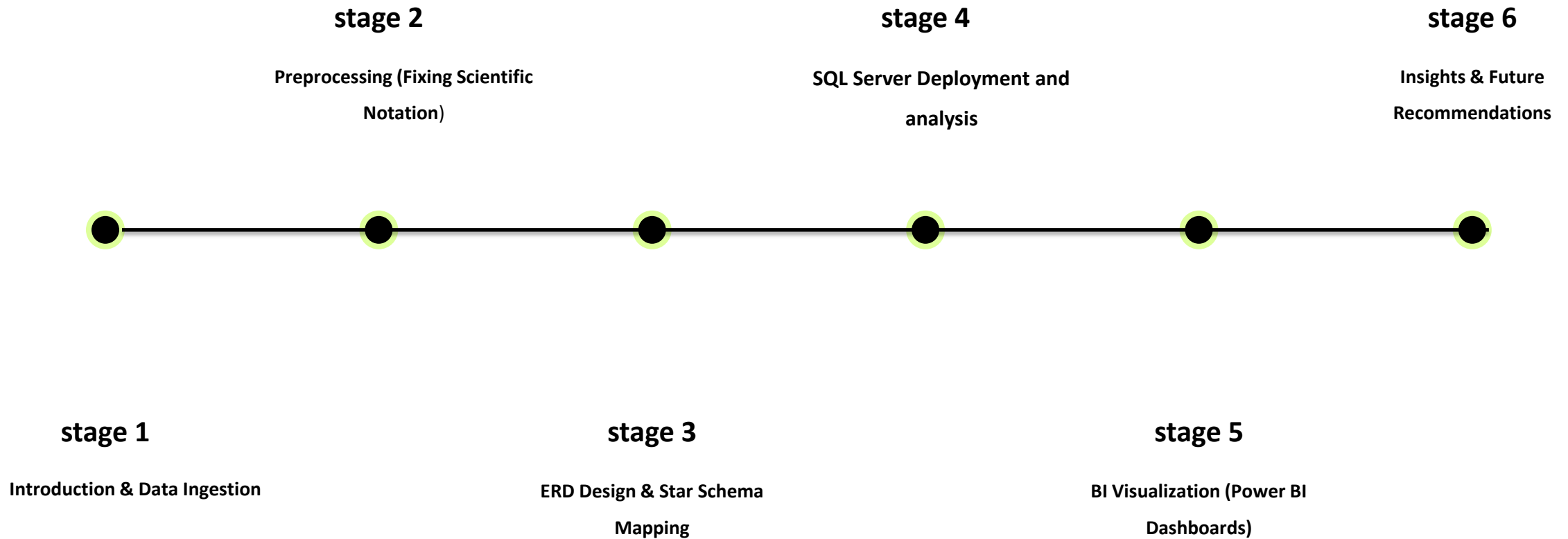
Global Energy Consumption Analysis



TEAM MEMBERS

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Project Timeline



The dataset provided contains detailed records of global energy consumption and production over a four-decade period from 1980 to 2021. The dataset includes information on energy consumption and production by country, region, and sector. It also includes information on energy prices, energy efficiency, and energy-related environmental indicators. The dataset is a valuable resource for researchers, policymakers, and analysts interested in global energy trends and challenges.



Satge 2: Preprocessing & Feature Engineering

1. Data Restructuring (Unpivoting)

Transformed the dataset by unpivoting 42 separate year columns (2021-1980) into a single 'record_year' and 'energy_value' column. This crucial step shifted the data from a wide format to a long format, enabling dynamic time-series analysis.

2. Formatting & Scientific Errors

Detected and resolved scientific notation issues (e.g., E- decroF .sVSC war eht ni (05 -12 tcirts a otni nmuloc 'eulav_ygrene' eht 12.%) tamrof taofl lamiced tniopf) to prevent data conversion crashes during SQL Server bulk insertion.

3. Handling Missing Values

Addressed inconsistent null representations in the raw files by replacing all occurrences of the string '--' with '-non decreoC .'0.0 gnisim gniniamer dellfi dna seulav ciremum etarucca erusne ot soresz htiw (sNaN) atad .snotiagergga lactiamehtam

4. Feature Engineering (Surrogate Keys)

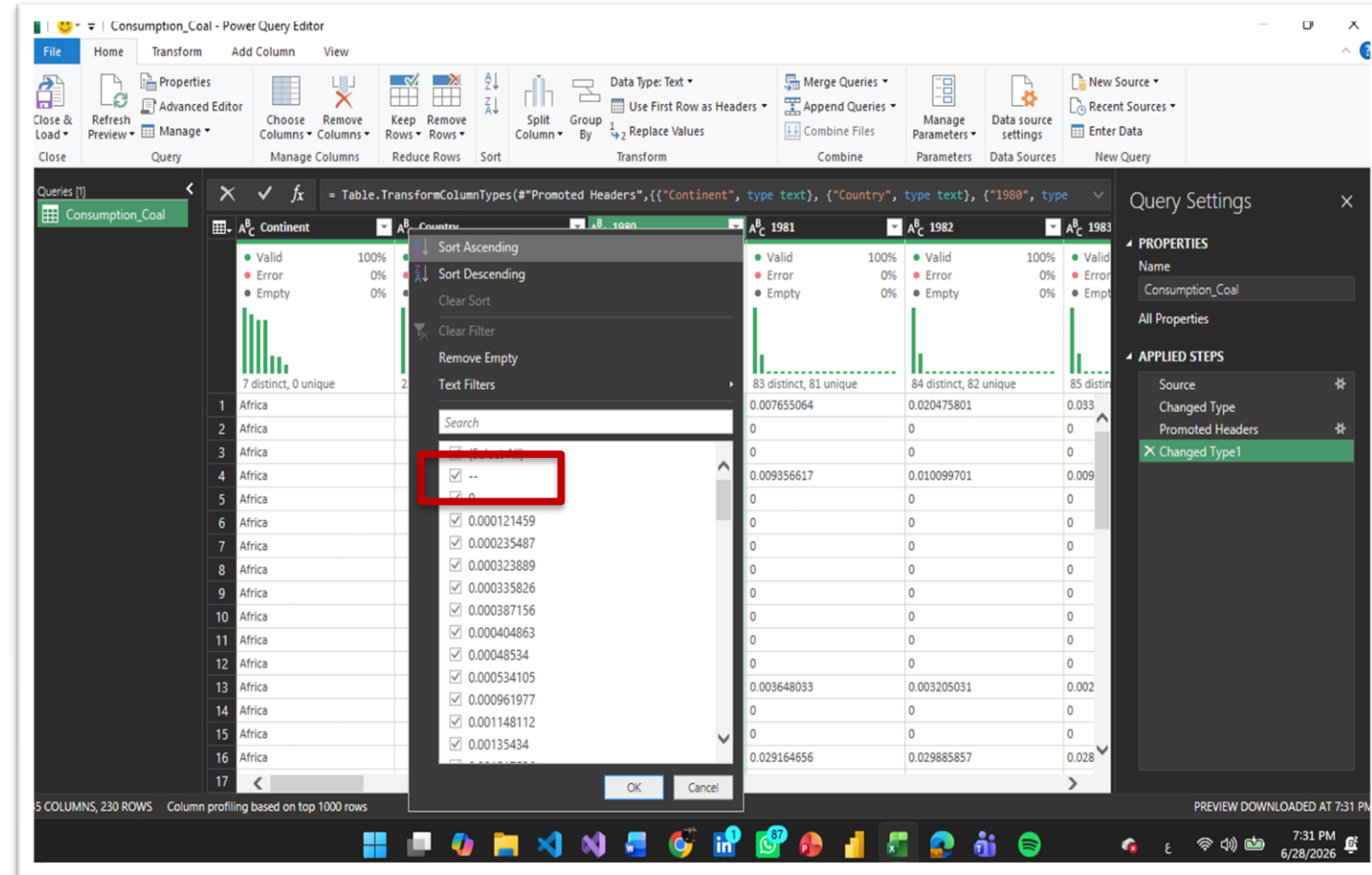
Created structured relational features by generating unique integer IDs for each dimension (continent_id, country_id, category_id, activity_id). Mapped these IDs back to the main dataset to build the central Fact table for the Star Schema.

Processing data

-Identifying Placeholders: Detected inconsistent null representations (the string "--") across numerical columns in the raw data.

-Data Cleansing: Replaced these text-based placeholders with numerical zeros (0.0
.yreuQ rewop gnisu

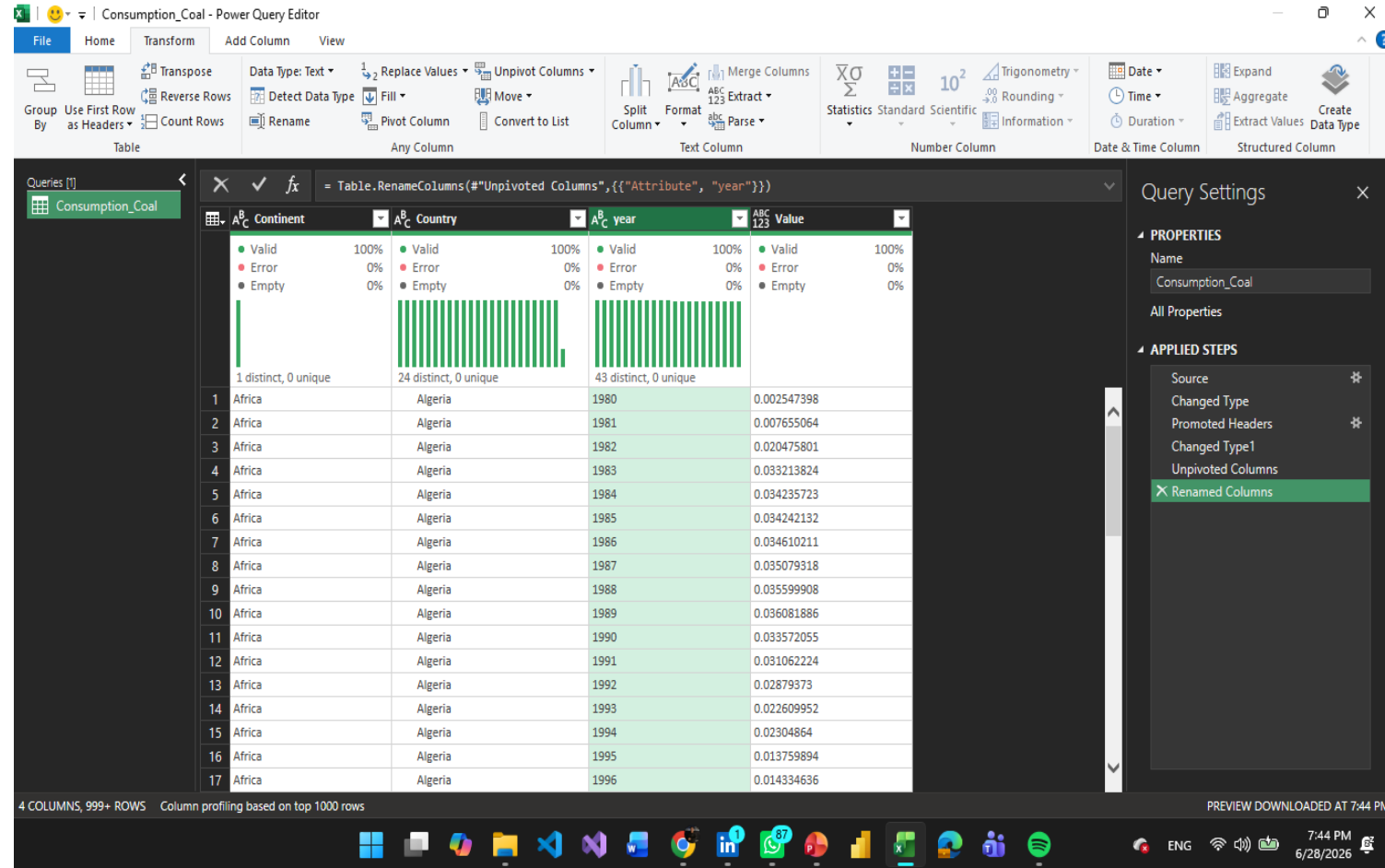
-Ensuring Accuracy: This vital step converts columns to strict numeric types, preventing crashes and errors during DAX calculations and visual aggregations.



Data Restructuring (Unpivoting)

Convert Wide Tables to Long Tables:
Converted 42 independent 'Year' columns into two flexible columns: 'Year' and 'Energy Value'. This dynamic transformation enables advanced time-series analysis and proper relational modeling.

Dependency Reduction: reduced column dependencies from 42 separate inputs to just two, simplifying logic and avoiding technical scientific notation issues.



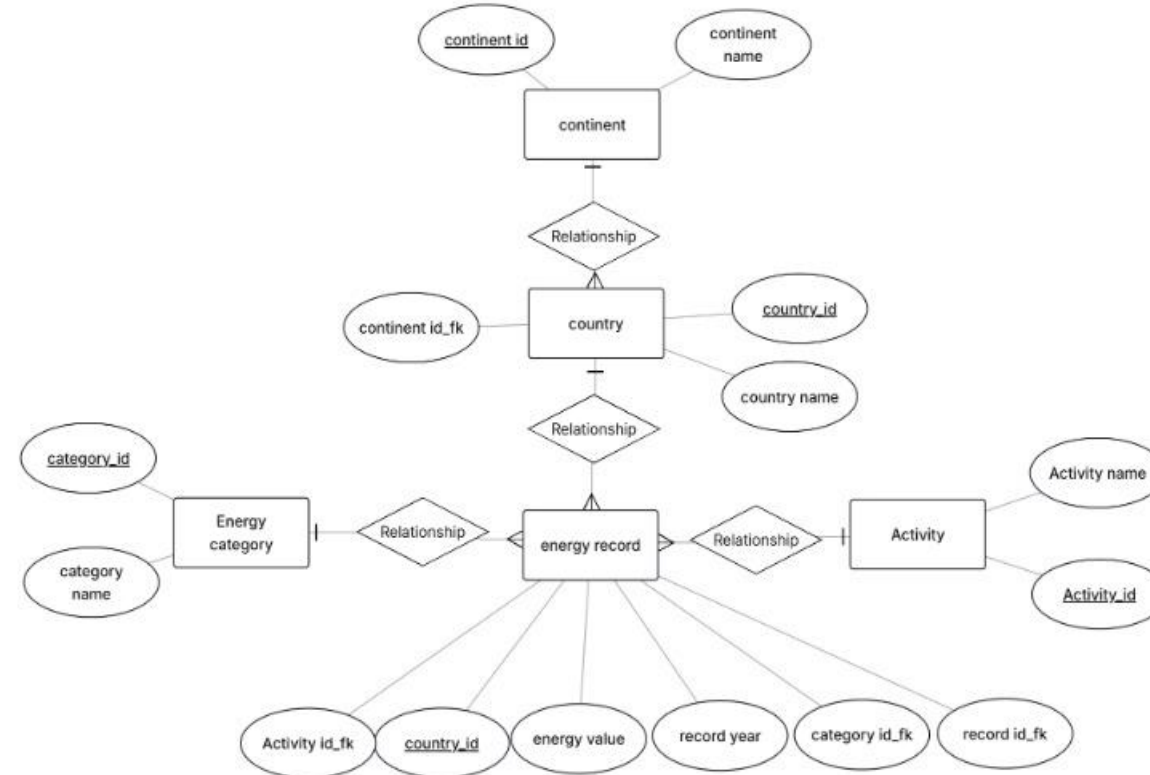
The screenshot displays the Power Query Editor interface for a query named 'Consumption_Coal'. The ribbon shows the 'Transform' tab with the 'Unpivot Columns' button highlighted. The data table is structured as follows:

Continent	Country	year	Value
Africa	Algeria	1980	0.002547398
Africa	Algeria	1981	0.007655064
Africa	Algeria	1982	0.020475801
Africa	Algeria	1983	0.033213824
Africa	Algeria	1984	0.034235723
Africa	Algeria	1985	0.034242132
Africa	Algeria	1986	0.034610211
Africa	Algeria	1987	0.035079318
Africa	Algeria	1988	0.035599908
Africa	Algeria	1989	0.036081886
Africa	Algeria	1990	0.033572055
Africa	Algeria	1991	0.031062224
Africa	Algeria	1992	0.02879373
Africa	Algeria	1993	0.022609952
Africa	Algeria	1994	0.02304864
Africa	Algeria	1995	0.013759894
Africa	Algeria	1996	0.014334636

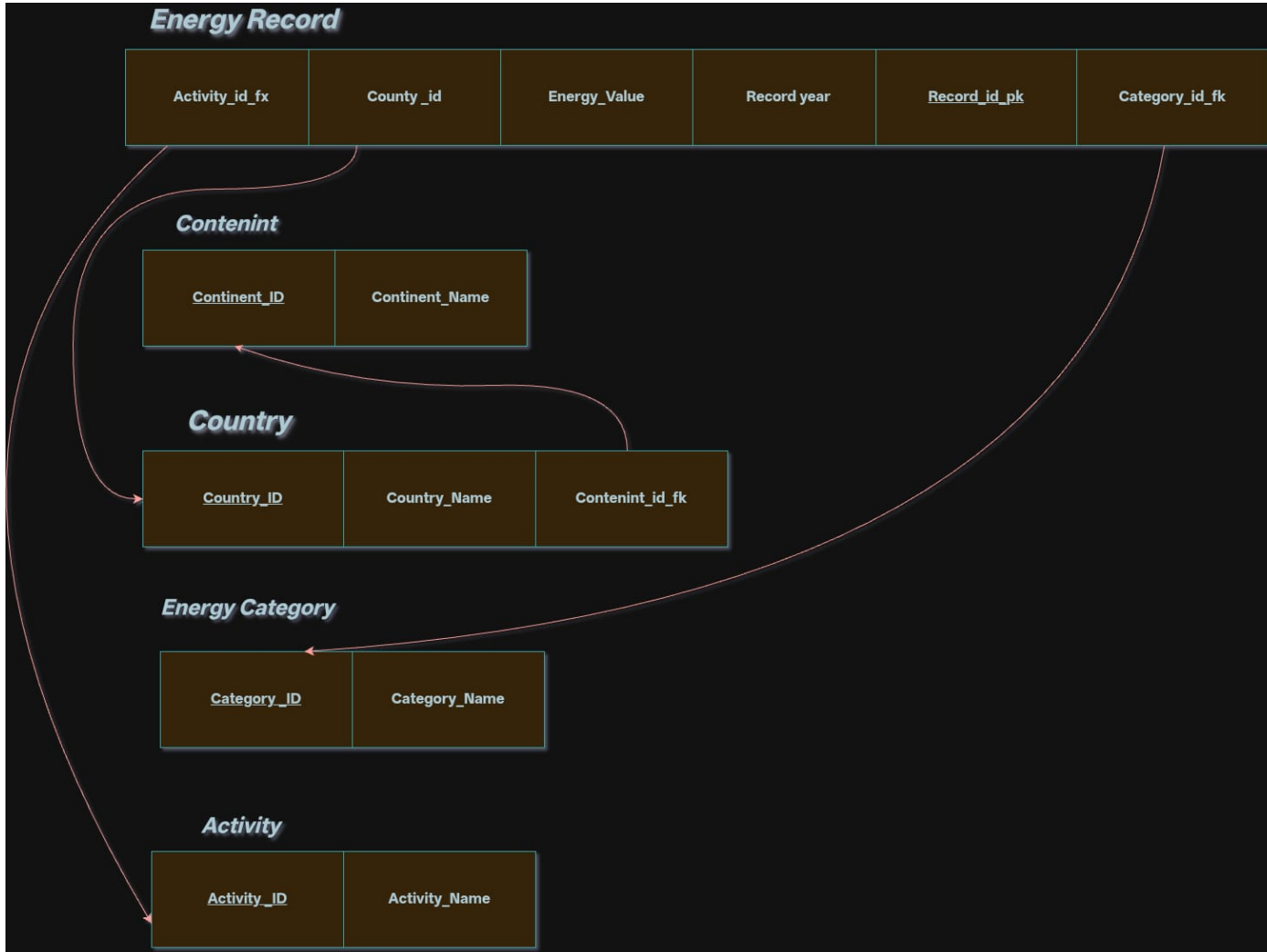
The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Unpivot Columns' and 'Renamed Columns'. The status bar at the bottom indicates '4 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

Stage3: Database Implementation & Data Ingestion (SQL)

Conceptual Entity-Relationship Diagram (ERD)



Normalized Data Model



Stage4:Database Implementation & Data Ingestion (SQL)

```

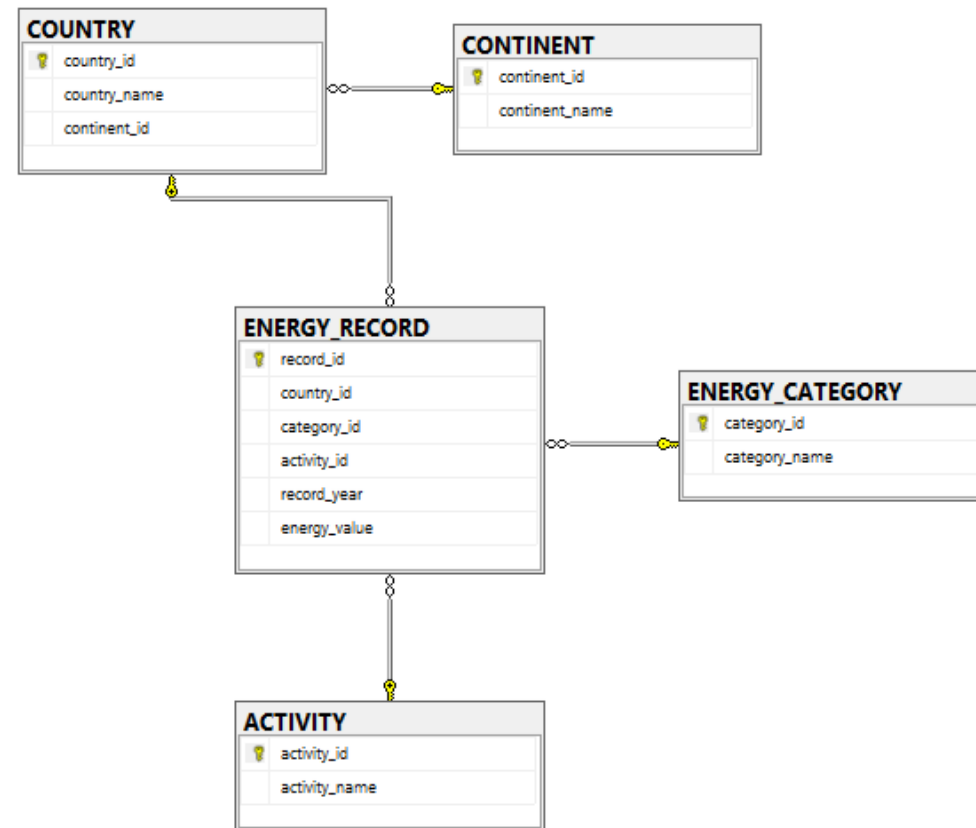
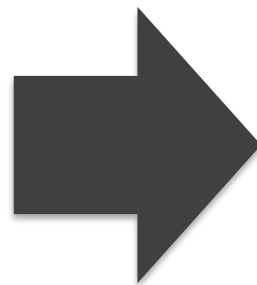
CREATE TABLE CONTINENT (
    continent_id INT PRIMARY KEY,
    continent_name VARCHAR(100) NOT NULL
);

CREATE TABLE COUNTRY (
    country_id INT PRIMARY KEY,
    country_name VARCHAR(100) NOT NULL,
    continent_id INT,
    FOREIGN KEY (continent_id) REFERENCES CONTINENT(continent_id)
);

CREATE TABLE ENERGY_CATEGORY (
    category_id INT PRIMARY KEY,
    category_name VARCHAR(100) NOT NULL
);

CREATE TABLE ACTIVITY (
    activity_id INT PRIMARY KEY,
    activity_name VARCHAR(50) NOT NULL
);

CREATE TABLE ENERGY_RECORD (
    record_id INT AUTO INCREMENT PRIMARY KEY,
    country_id INT,
    category_id INT,
    activity_id INT,
    record_year INT,
    energy_value DECIMAL(15, 4),
    FOREIGN KEY (country_id) REFERENCES COUNTRY(country_id),
    FOREIGN KEY (category_id) REFERENCES ENERGY_CATEGORY(category_id),
    FOREIGN KEY (activity_id) REFERENCES ACTIVITY(activity_id)
);
    
```



Stage4:Database Implementation & Data Ingestion (SQL)

SQL analysis

--What is the total global energy volume grouped by activity Production, Consumption?

- Which are the Top 5 countries with the highest overall energy production?

How is energy consumption distributed across different continents?

- What is the historical trend of global energy usage over the years?

--What is the total global energy consumption in the most recent year recorded? --Which energy category is the most dominant overall?

Answer important question

Stage4:Database Implementation & Data Ingestion (SQL)

SQL analysis

--What is the total global energy volume grouped by activity Production, Consumption?

```
SELECT A.activity_name,SUM(ER.energy_value) AS Total_Energy_Volume
FROM ENERGY_RECORD ER
JOIN ACTIVITY A ON ER.activity_id = A.activity_id
GROUP BY A.activity_name;
```

-- Which are the Top 5 countries with the highest overall energy production?

```
select top 5 c.country_name, sum(er.energy_value) as total_production
from energy_record er
join country c on er.country_id = c.country_id
join activity a on er.activity_id = a.activity_id
where a.activity_name = 'production'
group by c.country_name
order by total_production desc;
```

--How is energy consumption distributed across different continents?

```
select co.continent_name, sum(er.energy_value) as total_consumption
from energy_record er
join country c on er.country_id = c.country_id
join continent co on c.continent_id = co.continent_id
join activity a on er.activity_id = a.activity_id
where a.activity_name = 'consumption'
group by co.continent_name
order by total_consumption desc;
```

-- What is the historical trend of global energy usage over the years?

```
select record_year, sum(energy_value) as global_energy_volume
from energy_record
group by record_year
order by record_year asc;
```

--What is the total global energy consumption in the most recent year recorded?

```
select sum(er.energy_value) as latest_year_consumption
from energy_record er
join activity a on er.activity_id = a.activity_id
where a.activity_name = 'consumption'
and er.record_year = (select max(record_year) from energy_record);
```

--Which energy category is the most dominant overall?

```
select ec.category_name, sum(er.energy_value) as total_energy_value
from energy_record er
join energy_category ec on er.category_id = ec.category_id
group by ec.category_name
order by total_energy_value desc;
```

Stage 5: Visualization (Power BI) & Python(Plotly)

- **Plotly (Python)** for advanced, interactive data visualizations.
- **Overview of the core areas of analysis:**
 - Global Energy Trends (.sisylana (2021–1980
 - Continental and country-level distribution metrics.
 - Production versus consumption balance assessments.
 - Specialized deep-dive into nuclear energy production.
- **Impact & Value:** These performance insights enable stakeholders to identify long-term trends and regional energy dependencies, supporting data-driven decisions for future sustainability.



Global Energy Analysis



Overview

Petroleum Analysis

Oil Price analysis

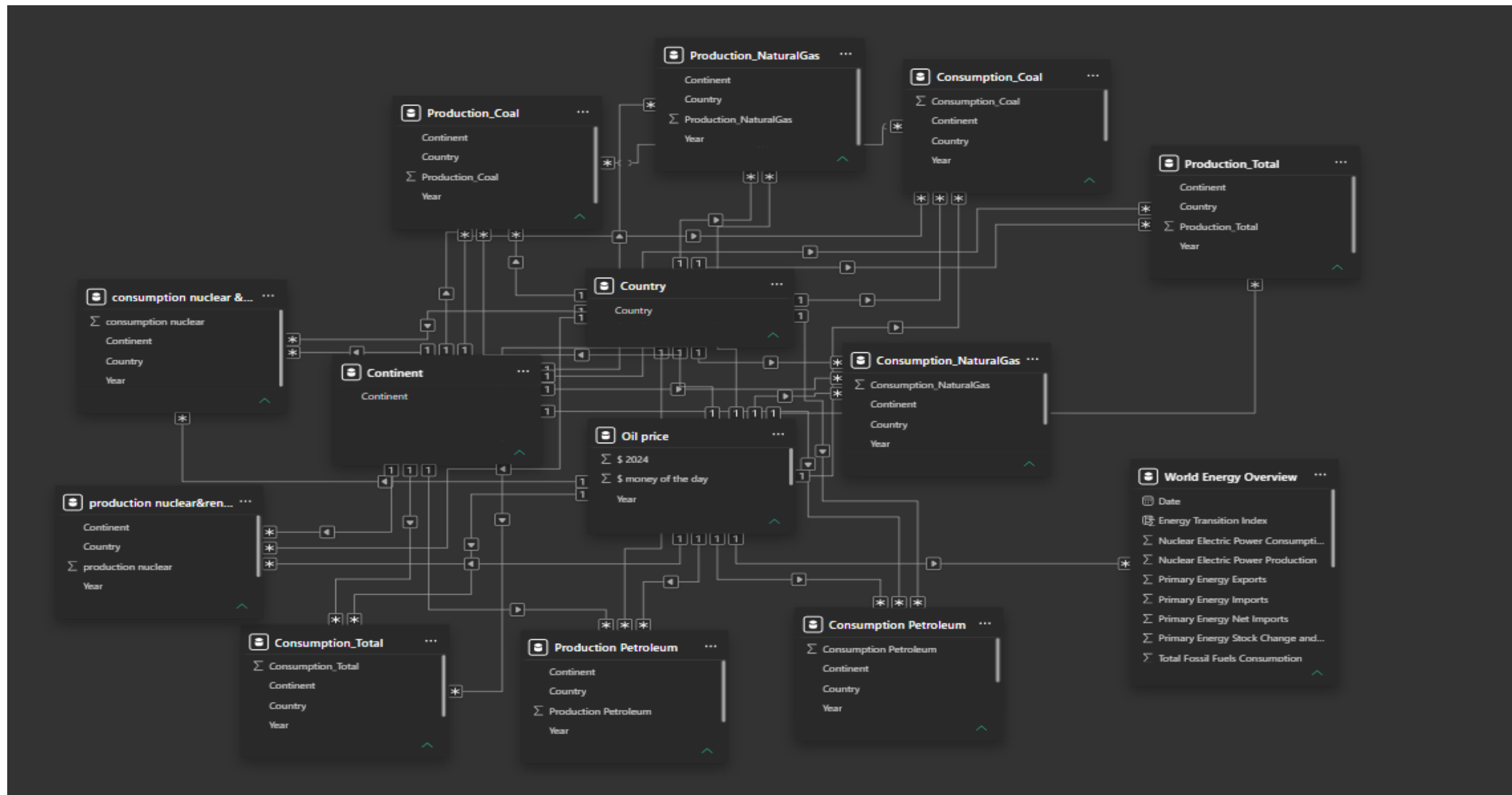
Renewable Energy

Nuclear Analysis

Coal Analysis

Natural Gas
Analysis





Global Energy Analysis



NEXT

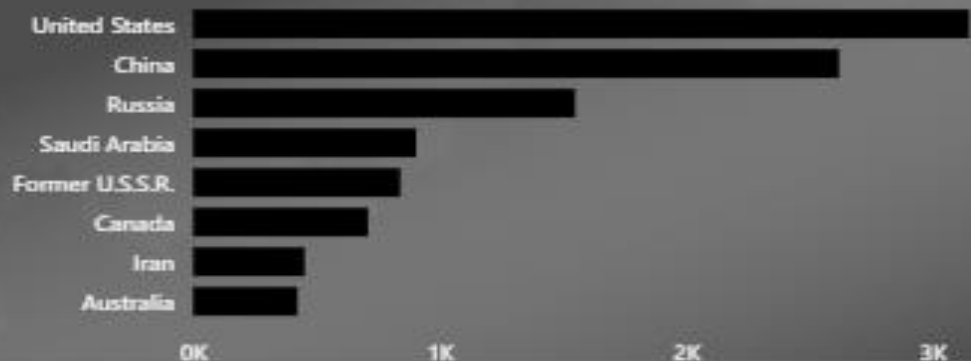
Total Consumption (Quads)

18.20K

Total Production (Quads)

18.16K

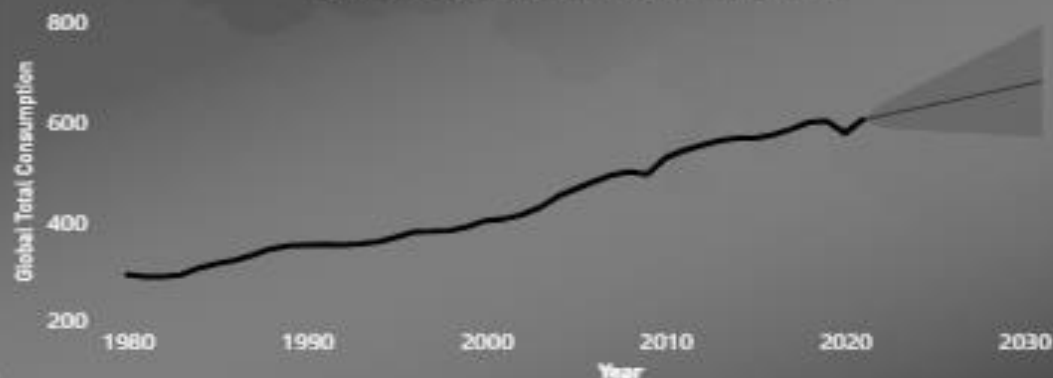
Total Production by Country



Total Consumption by Country



Global Total Consumption by Year



Year

1980	1983	1986	1989	1992	1995	1998
1981	1984	1987	1990	1993	1996	1999
1982	1985	1988	1991	1994	1997	2000

Petroleum Analysis


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petroleum consumption share

36.44%

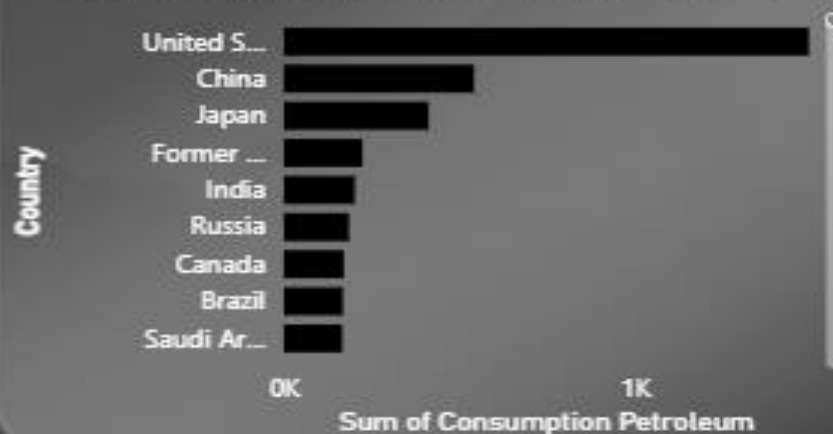
Total Petroleum Production

6.52K

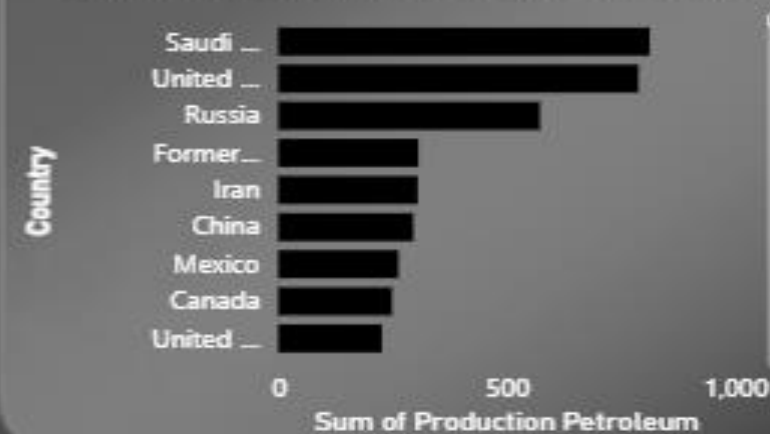
Sum of Consumption Petroleum

6.63K

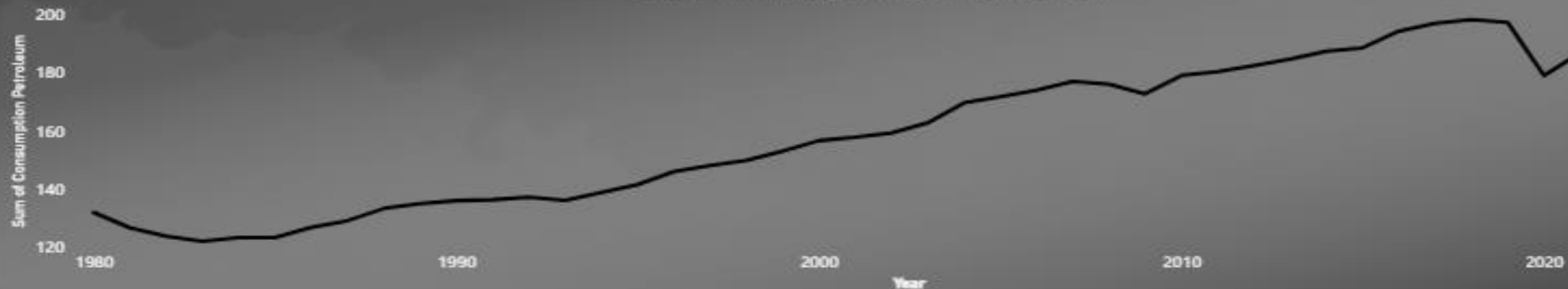
Sum of Consumption Petroleum by Country



Sum of Production Petroleum by Country



Sum of Consumption Petroleum by Year

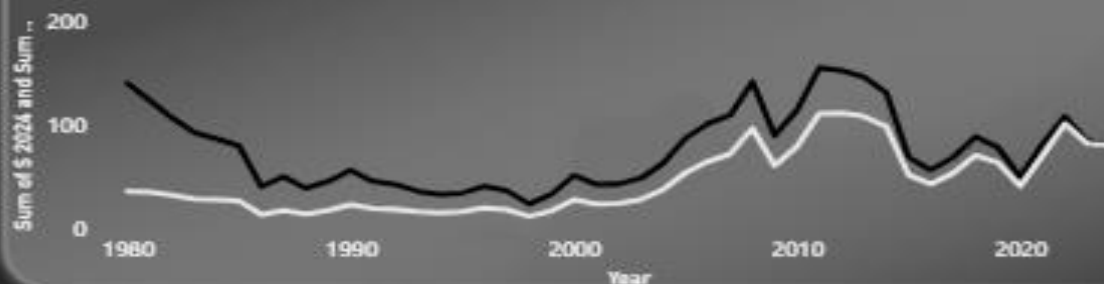


Oil Price Analysis


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Sum of \$ 2024 and Sum of \$ money of the day by Year

● Sum of \$ 2024 ● Sum of \$ money of the day



Sum of \$ 2024

3.45K

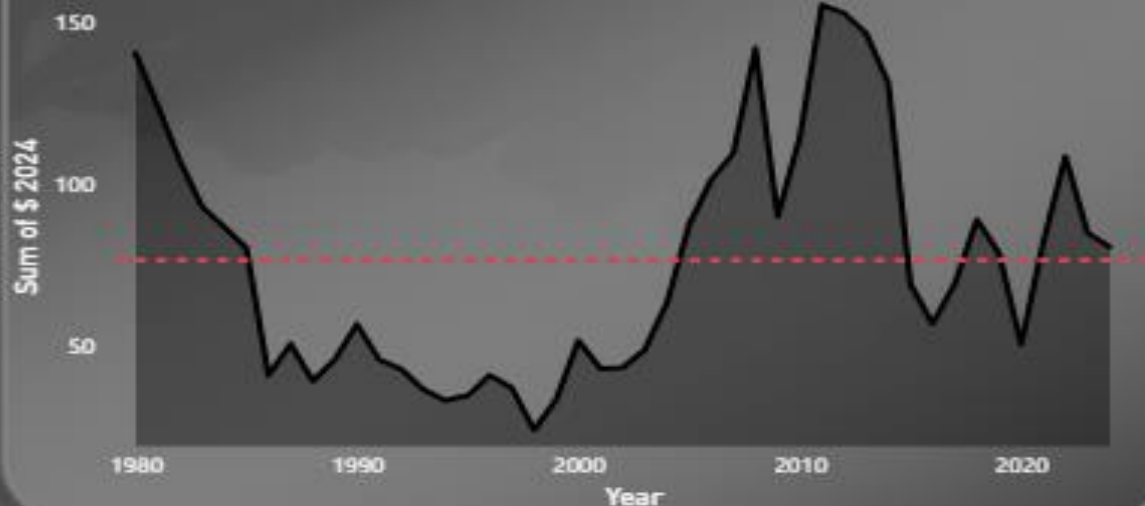
Average Oil Price

46.90

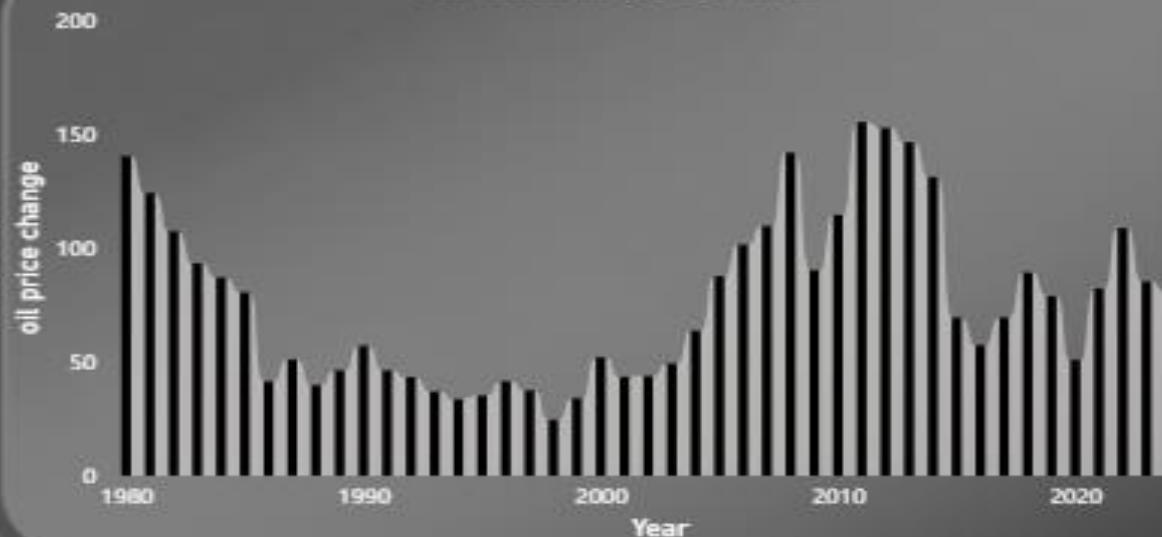
Year

- ☐ 1980
- ☐ 1981
- ☐ 1982
- ☐ 1983
- ☐ 1984

Sum of \$ 2024 by Year



oil price change by Year



Renewable Energy Analysis



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Carbon Intensity Index

67.86%

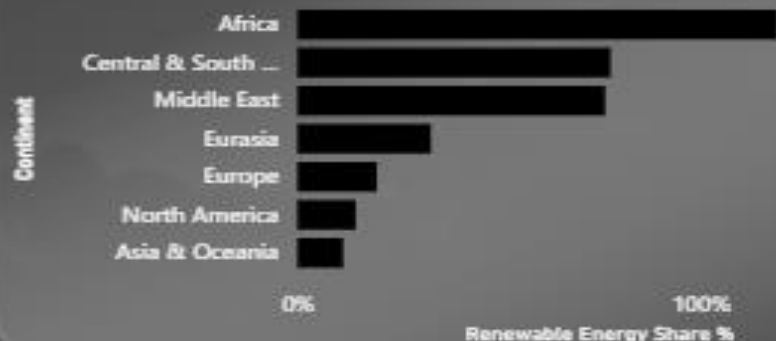
Clean to Dirty Ratio



Renewable Energy Share %

3.66%

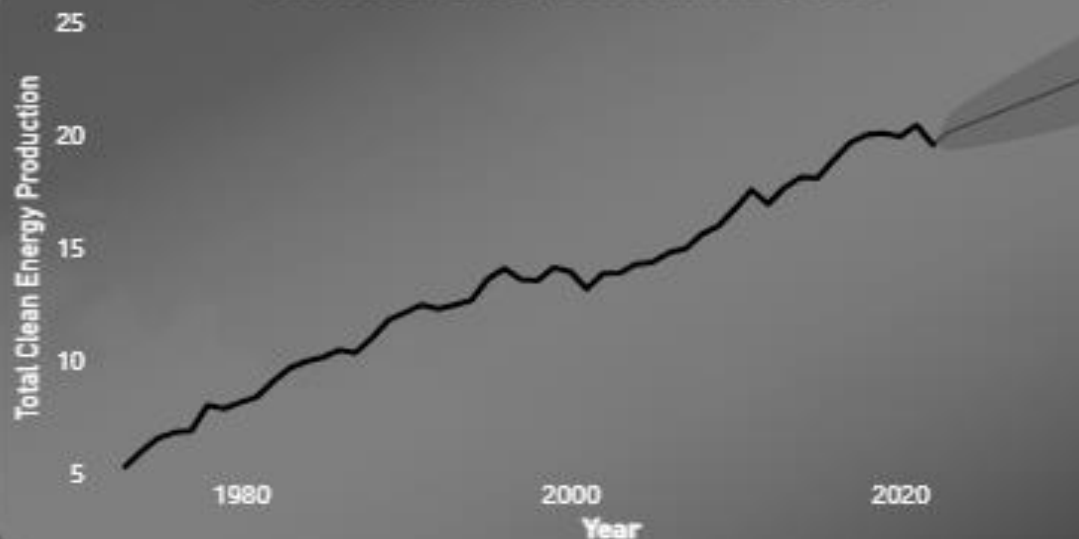
Renewable Energy Share % by Continent



Carbon Footprint Index by Country



Total Clean Energy Production by Year



Carbon Footprint Index	Country
2,526.53	China
2,461.06	United States
559.00	Japan
531.01	Russia
508.70	India
484.33	Former U.S.S.R.
285.47	Germany
248.57	United Kingdom
242.28	Canada
227.15	South Korea
187.17	France
185.21	Italy
179.38	Saudi Arabia
179.38	Saudi Arabia
12,351.33	

Nuclear Analysis


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Sum of consumption nuclear

2.52K

Nuclear Consumption Share

13.83%

Sum of production nuclear

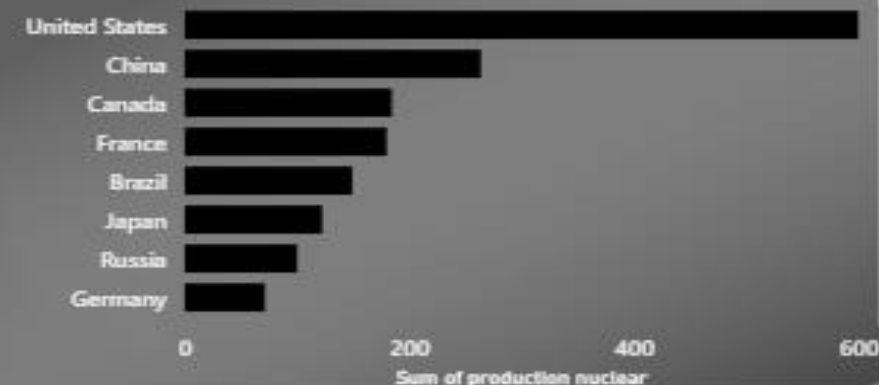
2.55K

Total Nuclear Electric Power Consumption

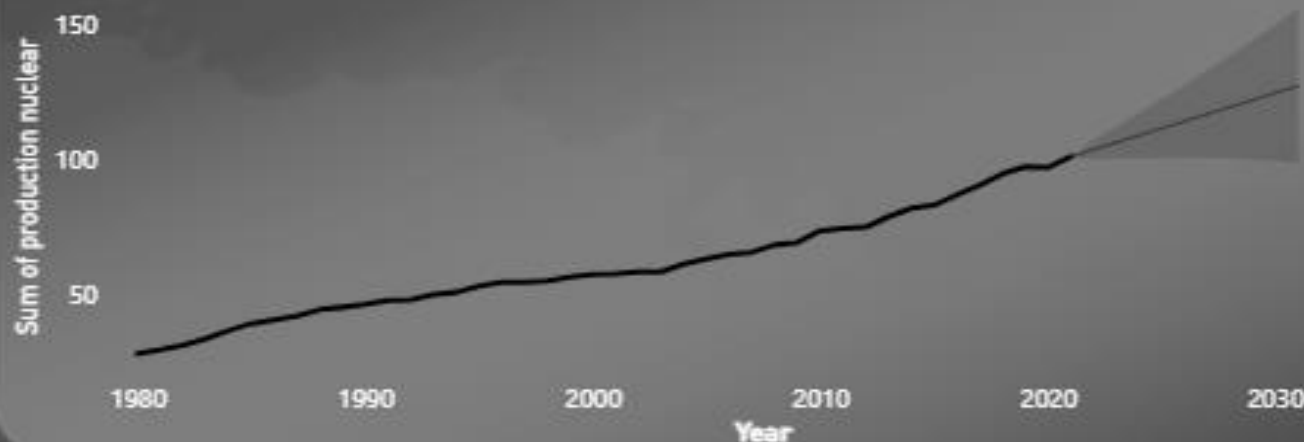
311.22

Sum of production nuclear by Country

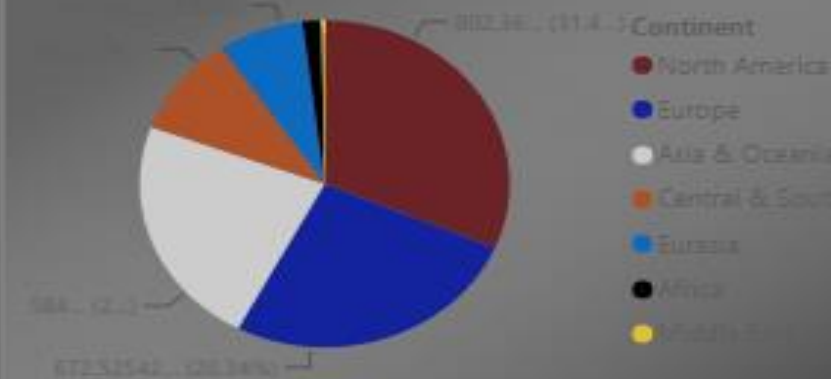
Country



Sum of production nuclear by Year



Sum of production nuclear by Continent



Coal Analysis



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Sum of _Coal Consumption

5.02K

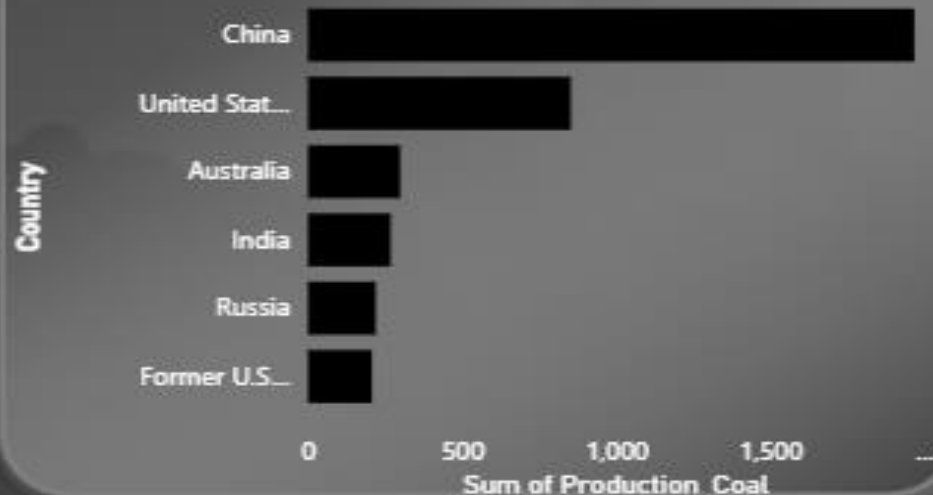
Sum of Coal Production

5.05K

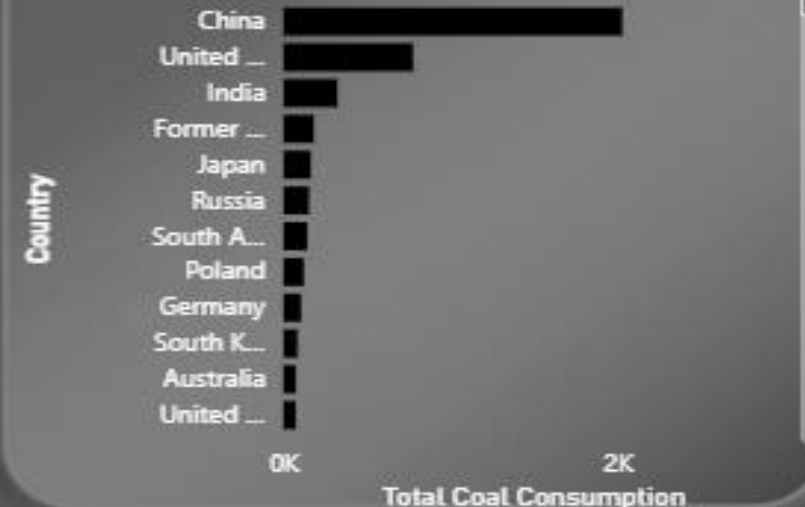
Coal Consumption Share

27.61%

Sum of Production_Coal by Country



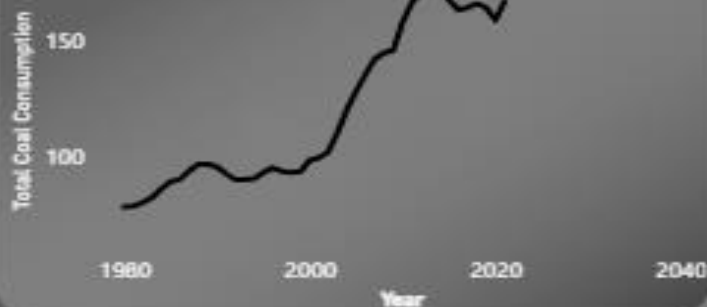
Total Coal Consumption by Country



Sum of Consumption_Coal by Continent



Total Coal Consumption by Year



Year

- ☐ 1980
- ☐ 1981
- ☐ 1982
- ☐ 1983
- ☐ 1984
- ☐ 1985

Natural Gas Analysis Dashboard



Sum of Production_NaturalGas

4.04K

Total Natural Gas Consumption

4.04K

Natural Gas Consumption Share

22.21%

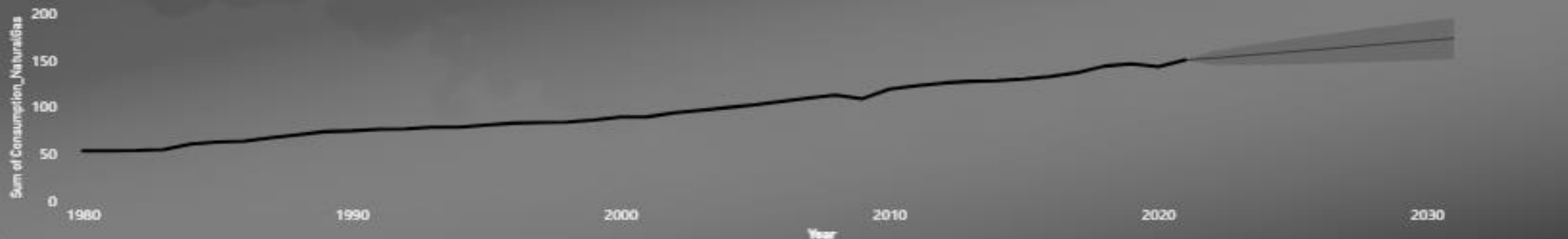
Sum of Consumption_NaturalGas by Country



Sum of Production_NaturalGas by Country



Sum of Consumption_NaturalGas by Year



Natural Gas Analysis Dashboard



BACK

Sum of Production_NaturalGas

4.04K

Total Natural Gas Consumption

4.04K

Natural Gas Consumption Share

22.21%

Sum of Consumption_NaturalGas by Country



Sum of Production_NaturalGas by Country



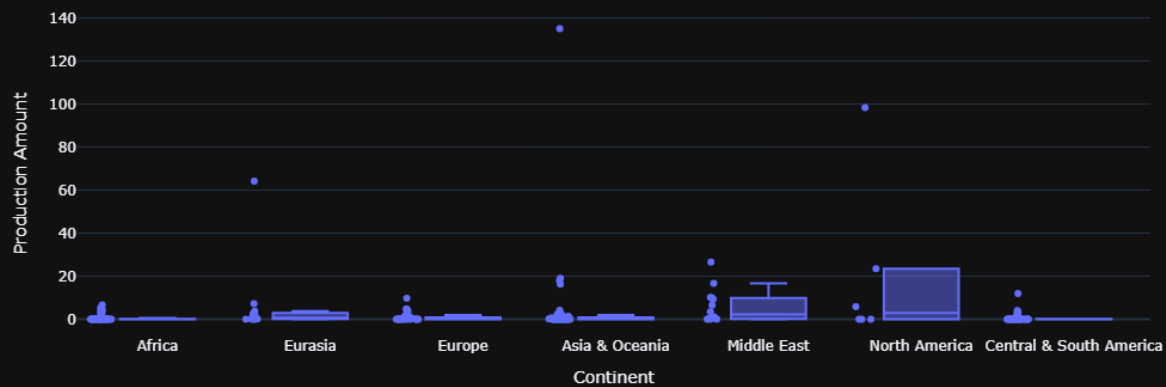
Sum of Consumption_NaturalGas by Year



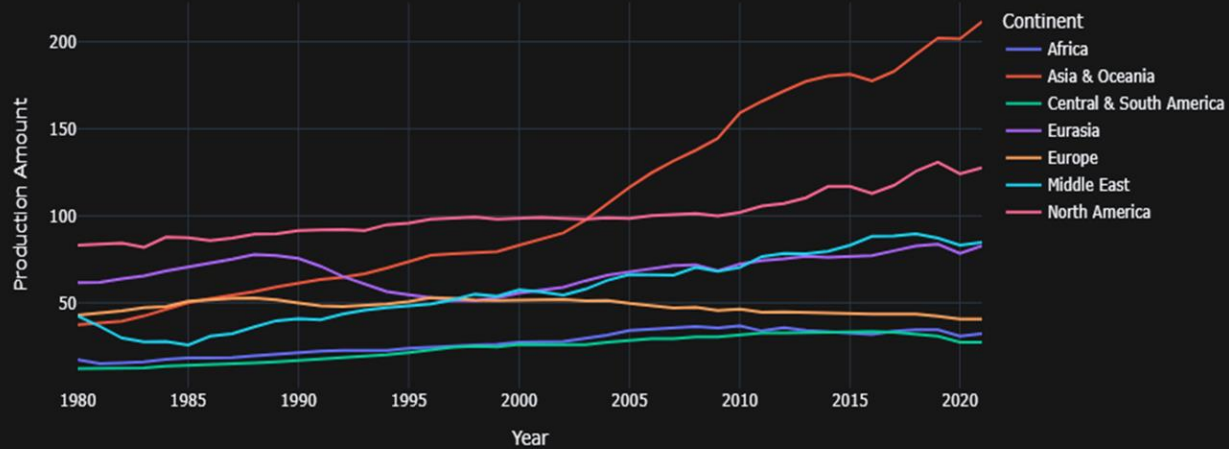
The Power of Plotly:

- **Interactive Insights:** Unlike static images, Plotly allows users to hover for precise values, zoom into specific timeframes, and pan across data points for deeper exploration.

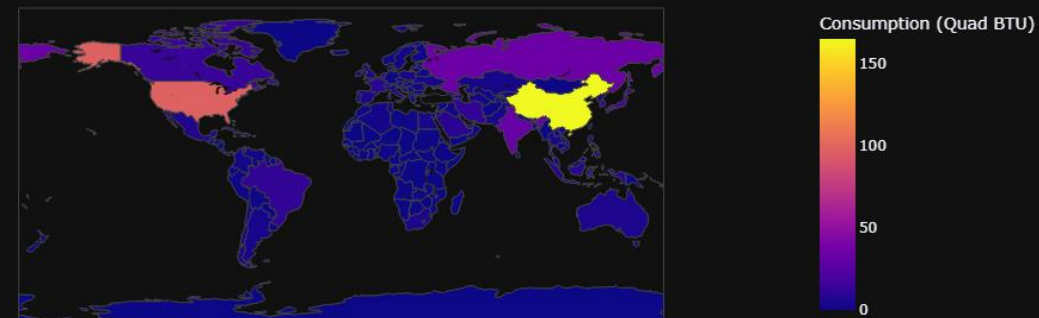
Production Distribution and Outliers by Continent (Year 2021)



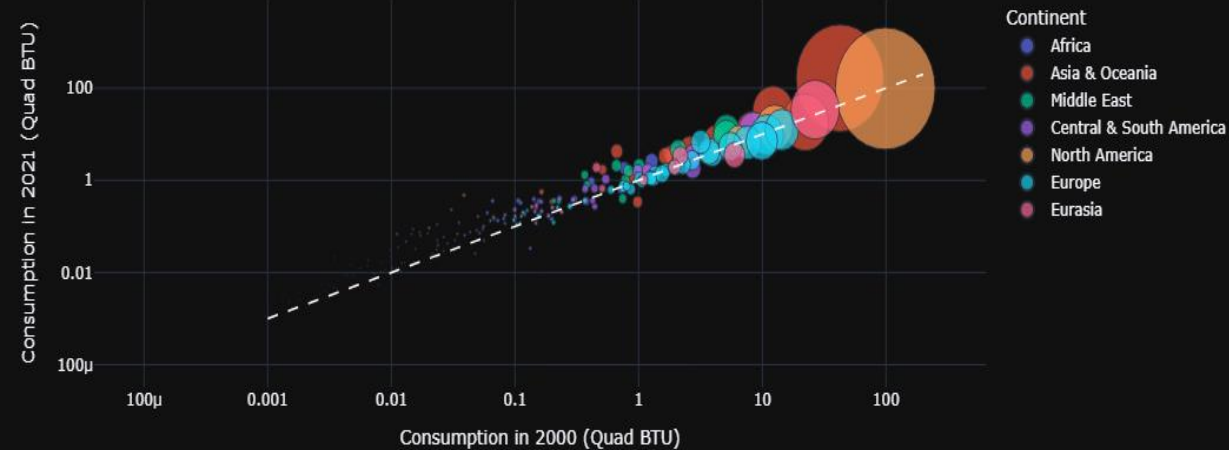
Total Production by Continent Over Time (1980-2021)



Global Energy Consumption Choropleth Map (2021)



Energy Consumption Growth: Year 2000 vs Year 2021 (Log Scale)



Project Summary

- Global Growth:** Total energy consumption around the world has shown a continuous and steady increase from 1980 to 2021.
- Top Consumers and Producers:** In 2021, China and the United States are the top consumers and producers of energy.
- Leading Region:** The Asia and Oceania continent takes the biggest piece of the pie in global energy use and production.
- Low Demand Regions:** Continents like Africa and Central and South America show very low energy consumption compared to the rest of the world.

Satge 6 :Insights & Future Recommendations

Key Insights

The Asian Industrial Boom: Over the last 20 years, Asia made a massive and rapid leap in energy demand. This reflects a huge shift in global factories and economies moving toward countries like China and India.

Stable vs Fast Growth: While North America and Europe are still heavy energy users, their growth curve is almost flat and stable. The real new demand is coming almost entirely from developing Asian markets.

The 2020 Global Drop: The timeline charts show a very clear dip in global energy use in the year 2020. This was due to the global pandemic, which led to a significant reduction in industrial and commercial energy consumption across all major economies.

The 1980 to 2021 Energy Transition: Over this period, there has been a significant shift in the global energy mix. While fossil fuels have remained a dominant source, there has been a growing emphasis on renewable energy sources and energy efficiency. This transition is driven by environmental concerns and the need for sustainable energy solutions.

Future Expectations

Continued Demand Growth: Global energy consumption will keep going up. As populations grow and new technologies like AI expand, the world will need much more power.

The Impact of Wars: Political conflicts in major energy regions, like Eurasia and the Middle East, will threaten global supplies. This means we can expect sudden shocks in oil and gas prices.

The Push for Energy Independence: Relying on imported fossil fuels is risky during political crises. Many countries will now focus on building local energy sources to protect their economies.

Faster Green Transition: The fear of supply cuts during wars will speed up the move to renewable and nuclear energy. Countries will invest heavily in these not just for the environment, but for their national security.

Data Cleaning

Challenge: Raw datasets contained missing values, inconsistent formats, and duplicates across 12 CSV files.

Solution: Used Power Query to standardize formats, remove duplicates, fill missing values, and create a unified data model.

Data Model Design

Challenge: Combining data from multiple files into a coherent structure with proper relationships.

Solution: Designed a star schema with fact and dimension tables, establishing one-to-many relationships for optimal performance.

DAX Measures

Challenge: Writing complex calculations for KPIs, shares, and forecasting while maintaining performance.

Solution: Created optimized DAX measures with variables and filter context to calculate KPIs efficiently.

Dashboard Performance

Challenge: Large dataset caused slow loading times and lag during interactions.

Solution: Optimized the data model, reduced cardinality, and used aggregation tables to improve response time.

Team Collaboration

Challenge: Coordinating work among 6 team members on the same project files.

Solution: Used GitHub for version control, organized files in repositories, and maintained clear documentation.

Visualization Choice

Challenge: Selecting the most appropriate charts to represent different types of data clearly.

Solution: Researched best practices and chose bar charts for comparisons, line charts for trends, and pie charts for shares.

FUTURE

01 Real-Time Data Integration

Integrate live energy data from online APIs for up-to-date insights.

02 Forecasting Models

Add advanced forecasting models to predict future energy consumption trends.

03 Power BI Service

Publish the dashboard to Power BI Service for online access and sharing.

04 Expanded Coverage

Include more countries and additional energy-related indicators.

05 Advanced Filtering

Add drill-through pages and advanced filtering for deeper analysis.

06 Performance Optimization

Optimize data model and DAX measures for faster dashboard performance.

07 Row-Level Security

Implement RLS to support secure access for different user roles.

08 Mobile Layout

Create a mobile-friendly dashboard layout for smartphones and tablets.

RESULTS

The project successfully transformed raw energy consumption data into an interactive and insightful Power BI dashboard. The dashboard enables users to explore global energy trends through dynamic visualizations and filters.

7

Dashboard Pages

Interactive pages covering all energy sources

12

CSV Files

Raw datasets cleaned and transformed

40+

Years of Data

From 1980 to present with forecasting to 2030

5

Energy Sources

Petroleum, Coal, Natural Gas, Nuclear, Renewable

6

Team Members

Collaborated using GitHub version control

∞

Insights

Data-driven decisions for energy planning

Project Achievements

- Developed an interactive dashboard for analyzing global energy consumption
- Successfully cleaned and transformed the dataset using Power Query
- Created DAX measures to calculate key performance indicators (KPIs)
- Enabled dynamic filtering by country and year for detailed analysis
- Compared different energy sources with clear visualizations
- Improved data accessibility by presenting complex information in a user-friendly format

Thank you